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SURVEY

A Survey on Integration of Empathy Cues Into Empathetic Text Generation

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ABSTRACT In recent years, affective computing has seen a surge in empathetic response generation algorithms. These incorporate different empathy-related concepts to increase their perceived behavioral empathy. This survey aims to define the predominant research directions and offer structure to the field through the lens of empathy cues. In this definition, we include the concepts, factors, and components of interpersonal communication that relate to the expression of empathy. We describe the four general methods taken for extracting empathy cues from multiple sources. Subsequently, we explore three major strategies used to incorporate these cues into the text generation pipeline. Then, we present the methods used by the most prominent algorithms for evaluating their results and present a novel hierarchical view of performance across the field. Finally, we present areas of opportunity for novel approaches.

INDEX TERMS Affective computing, emotion, empathetic text generation, empathy, natural language processing, text generation.

I. INTRODUCTION

The presence of affective computing in industry and academia has increased significantly between 2019 and 2025 [1]. In addition, conversational systems have become more common since the appearance of Large Language Models (LLMs) like ChatGPT, which aim for human-like communication [2]. These factors make empathy an intriguing avenue for research, especially in conversational systems [3], as it can help increase factors like perceived intelligence, human likeness, and communication ability in conversational systems [4], [5]. As such, empathetic text generation (ETG) has become a popular task in this field. However, the avenues pursued for ETG are currently disorganized. They use various concepts, definitions, architectures, and sources that make it difficult to obtain a clear view of the development trends and how the components of empathy can be used for this task.

In psychological literature, empathy is described as a concept with multiple components or dimensions [6], [7], [8], [9], [10]. This description of empathy provides a psychological basis for defining it as a trait or state comprised of multiple phenomena like actions, thoughts, emotions, and intents. In affective computing, behavioral empathy [6] refers to phenomena that can be observed, measured, and incorporated into computer systems to mimic or detect empathetic behavior [11]. In this review, we explore how this concept allows researchers to integrate phenomena related to empathy into ETG systems.

We refer to any element that can be used to contribute to empathetic behavior as **empathy cues**. These are attributes, factors, values, or concepts related to a dimension of empathy. For example, emotions, polarity, or mimicry [11]. Integrating these cues has been shown to increase the perceived empathy

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ability of conversational systems [12]. Therefore, we are interested in the methodologies for extracting and integrating them into text generation applications. Likewise, we aim to explore their effects in performance.

Empathetic conversational systems have been increasingly explored in recent years. In particular, ETG approaches have exploded since the publication of the EMPATHETICDIALOGUES dataset [13], as observed by Raamkumar and Yang [12] in a previous survey. These approaches obtain good performance by implementing some empathy cues into their architectures. For example, emotions [14], [15], intent [16], and communication mechanisms [17]. However, the diversity in their methodologies makes it very difficult to identify and catalog the nature of these cues, as well as their sources. As such, researchers are left to wonder which cues are useful and how they can be leveraged. In this survey, we aim to obtain an analysis of ETG that empowers affective computing researchers with the tools to adopt empathy cues in their dialogue systems.

Our first objective consists of identifying different methods used to obtain empathy cues. This was because the diversity in methodologies and scattered literature makes it difficult for researchers to select and obtain variables that can contribute to empathy. We explore the state-of-the-art (SOTA) and identify 4 broad categories used to extract empathy cues from various sources. We describe each method and explore the types of cues that can be obtained through it. This will allow future researchers to identify sources and types of cues for their proposed ETG systems. A methodology used to obtain empathy cues from text, either involving human judgment or automatic, is referred to as **sourcing method** in this survey.

Another objective for this survey is to explore the methods used to add empathy cues into the text generation process. We categorize the ETG systems from our survey into 3 strategies. These are based on how an empathy cue is leveraged to improve the perceived empathy of the responses. We refer to these methodologies as **integration strategies**. We describe how they are used in the SOTA, the type of cues used with each strategy, and their characteristics. This was carried out to identify trends and allow ETG systems to build upon our findings.

Finally, we were interested in how ETG systems are evaluated and whether we could obtain an overview of performance across the SOTA. We provide a novel framework and an overview of the field. Likewise, we identify common metrics and human evaluation strategies.

This survey aims to empower researchers with the tools to identify, extract, and leverage empathy cues for ETG systems. Therefore, this survey can serve as a starting point for developing novel ETG systems that use empathy cues.

Our survey is presented as follows: First, we present the definition of empathy we adopted in Section II. Next, Section III presents our ETG framework, with different elements that we discuss further in the survey. Section IV describes some applications for empathetic text. Then, Section V presents the datasets used in ETG. Subsequently, Section VI describes the sourcing methods, along with their trends and cues extracted by each one. Following that, Section VII explores the integration strategies along with their corresponding sourcing methods, and how the trends have changed over time. We provide an overview of evaluation of ETG systems in Section VIII. Finally, we discuss our findings and present areas of opportunity in Section IX.

A. SCOPE OF THIS SURVEY

Empathy and empathetic dialogue generation is an extensive topic that can be explored through different lenses. For our survey, we focus on ETG systems that incorporate empathy cues to influence the judged empathy [6] of texts from 2019 to 2025, as this covers the time frame since EMPATHETICDIALOGUES (ED) [13] has been publicly available. This dataset has laid the foundation for ETG and is one of the most important in the field [12], [18]. We discuss more about this dataset in Section V. Furthermore, we will focus on approaches that state the empathy cues used. These articles must also present the mechanism for leveraging these components to create empathetic text.

We conducted the selection process by searching IEEE Xplore and Google Scholar for relevant articles for our survey. Our search criteria consisted of the terms *empathetic text*, *empathetic dialogues*, *empathy generation*, *empathetic text generation*, *empathetic chatbots*, *empathetic conversation generation*, *empathy in chatbots*, and *empathetic response generation*. Our sorting criterion was of relevance and date. This search allowed us to identify 502 articles.

We removed articles on the field of psychology and sociology. Therefore, our review focused on computer science and affective computing. This was done through metadata by discarding journals and conferences unrelated to computing. Likewise, we manually discarded papers using their titles. This left us with 367 articles related to empathetic response generation. We then applied our selection criteria by reading both the title and abstract.

The selection criteria was to discard articles focused on empathetic dialogues using multi-modal input, including audio or video. We removed those approaches that did not incorporate empathy cues or did not disclose their databases or mathematical mechanisms. In addition, we focus exclusively on text generation in one-on-one open-domain dialogue. This is because previous research has shown ethical concerns with artificial empathy in healthcare, as well as condemning the use of it as a manipulation tactic [19]. As such, we remove papers on empathy generation for healthcare, therapy, and customer service.

Finally, we searched the references of our relevant articles to find the work that they were based on. This allowed us to obtain the source of their respective empathy components, construct a timeline, and organize the articles by their integration strategy and their sourcing methods. The final

articles covered in depth in our review were 45. We also include two relevant antecedents [20], [21]. We present a summary of all models of interest in Table 5.

Previous surveys have explored ETG using different frameworks [3], [12], [22], [23], [24], [25]. We focus on examining the methodology for obtaining and integrating empathy cues in text generation pipelines. Likewise, we explore the effects, nature, and usefulness of these cues to influence the perceived empathy of text responses.

B. CONTRIBUTIONS

We present a conceptual framework for the task of empathetic text generation using the concept of empathy cues: Empathy cues for text generation (EmpCue4TG). We consider empathy cues any concept, factor, value, or indicator that can contribute to identify and indicate empathy in textual communication, besides the utterance. Thanks to our conceptual framework, we obtain a method to explore the task of using empathy cues in text generation with multiple core algorithms. Our contributions are as follows:

1) We provide an analysis of sourcing methods used to obtain empathy cues in ETG systems. Four methods were identified: **manual annotation**, **retrieval**, **classification**, and **generation models**. We explore the nature of cues extracted using each sourcing method. Likewise, categorize each ETG system according to their sourcing method. Then, we describe the trends, advantages, and disadvantages of each. This allows researchers to identify, replicate, and analyze the diverse cues used in ETG and provides a frame of reference for extracting empathy cues for text interactions.
2) We organize the ETG task according to the strategies used to integrate empathy cues. Therefore, we create a frame of reference based on the use of empathy cues for improving the performance of ETG system. Three integration strategies were identified: **context augmentation**, **variable conditioning**, or a **hybrid strategy**. Besides, we describe which empathy cues are compatible with each integration strategy. Likewise, we identify trends in development through time. This exploration can serve as guidelines for future research.
3) Finally, we describe how to manually and automatically evaluate ETG systems. Then, we provide an analysis of the SOTA performance, baselines, and comparisons between algorithms to provide future research with a benchmark selection method.

II. EMPATHY DEFINITION

In this article, we adopt the definition of empathy proposed by Clark et al. [6]. It was developed based on comprehensive reviews of the concept [7] and it has precedent in affective computing applications [11].

Empathy is a construct with three main dimensions: (1) affective, (2) cognitive, and (3) behavioral. (1) Affective empathy refers to the state or trait of matching emotions with another person. (2) Cognitive empathy refers to the state or trait of understanding the internal state of another person. Finally, (3) behavioral empathy refers to verbal and non-verbal behaviors that demonstrate affective and cognitive empathy, including behavioral mirroring and empathetic communication [6].

The main reason for adopting this definition is the inclusion of the behavioral dimension. ETG systems are incapable of experiencing internal empathy [19]. However, behavioral empathy allows us to describe artificial agents as displaying a real dimension of empathy in the form of observable behaviors. Particularly, empathetic communication. Other definitions like the one proposed by Batson [10], are predominantly focused on the internal processes that make up empathy, with few phenomena being observable. Additionally, Clark et al. state that there is a research gap regarding the nature of elements that make up empathetic behavior. As such, our aim is to explore these in the context of ETG.

We are aware that there are multiple works with different definitions of empathy in affective computing [6], [7], [9], [26], [27], [28], [29]. However, we decided to use a different definition. This was because most were proposed when the field was still nascent [22]. As such, their empathy models lack the advantages of concepts like behavioral empathy [9], [27], [28]. Furthermore, those definitions are a better fit for applications different from ETG, such as the work by Brannstromm et al. [9] and Yalçin et al. [28], which can be great for studying empathy performance in chatbots, but whose elements can be difficult to adapt to the generation process.

III. EMPATHY CUES FOR TEXT GENERATION

Empathetic response generation is the task of enabling machines to understand and respond to emotions and feelings [18]. Empathetic Text Generation (ETG) is a subtask focused on text responses. We apply the empathy definition in Section II to obtain a conceptual framework for this task.

Our definition of empathy is based on affective, cognitive, and behavioral dimensions. We focus on the last dimension for ETG, as it is composed of observable elements related to the other two dimensions. These can be replicated by artificial agents. In our framework, ETG is a task in affective computing that aims to replicate behavioral empathy using text responses. Many elements and concepts related to empathy can be used for this purpose.

We present the EmpCue4TG framework in Figure 1. Under this conceptual framework, a text generation algorithm is trained on a dataset to produce responses given an input. However, this process is supplemented by using empathy cues. For example, emotion [30], emotion-cause [31], or communication mechanisms [32]. We present a basic text generation pipeline with the following elements: input, output, and core text generation algorithm. The core text generation algorithm can vary from transformers, to LLMs.

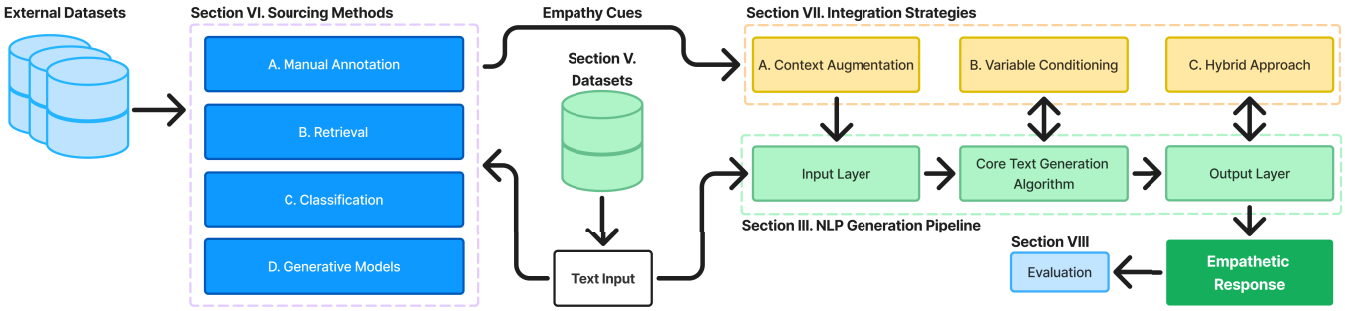


FIGURE 1. Conceptual framework for generating empathetic text. We display the survey section for each module.

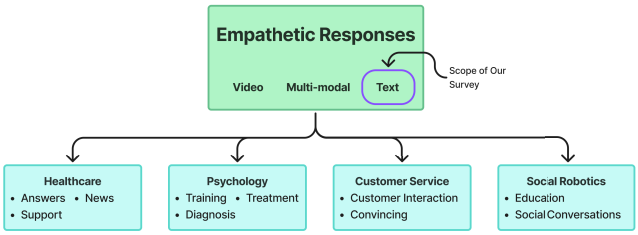


FIGURE 2. Applications examples.

In this manner, we accommodate the diverse methodologies used for text generation. This conceptual framework allows us to categorize ETG systems according to the methods for extracting and integrating empathy cues.

Empathy cues are extracted from a text utterance before it is sent to the input layer of the pipeline. To do this, a sourcing method is applied: manual annotation, classification, retrieval, and generative models. These method use external models, datasets and human judgment for this task. Cues are then are added to the NLP pipeline using one of these integration strategies: context augmentation, variable conditioning, or a hybrid strategy. Subsequently, the text input and cues are processed and the system provides an empathetic response. Then, it is evaluated according to human and automatic metrics.

We explore the ETG systems included in our survey using EmpCue4TG. Under this framing, the goal of using empathy cues is to influence a text generation algorithm to make text responses more empathetic. We will explore the framework elements related to these cues in the following sections of this survey.

IV. APPLICATIONS

Empathetic response generation has many applications beyond one-on-one open domain conversations [12], [33]. In the field of affective computing, we identified four main areas for applying artificial empathy: healthcare, psychology, customer service, and social robotics. These areas can be explored through text, video, or multi-modal formats in diverse scenarios. We present an overview of the empathetic responses task and highlight ETG in Figure 2.

Empathetic agents can be beneficial in healthcare for tasks such as: answering patient questions [5], [34], [35], discussing sensitive healthcare topics [36], or informing patients of difficult news [37]. As such, they can be useful for professionals in this field to supplement the care of patients while considering their emotional needs [5]. However, this application has been criticized on the grounds that clinical empathy requires internal mental states and a genuine helping intention to provide effective care [19]. Furthermore, artificial clinical empathy can be considered a form of deception or manipulation [38].

Psychological support has been explored as an application of empathetic computing. For example, empathetic artificial agents have been suggested as training tools for therapists [32], [39]. Likewise, these tools have been proposed for diagnosing and treating mental health conditions [22], [40], [41]. Nevertheless, this field must be further scrutinized and improved, as empathetic care from artificial agents has been shown to backfire [42].

Empathy in customer service has been explored in previous studies which have shown that including this characteristic leads to beneficial effects. For example, increasing the quality of social experience for customers and encouraging acceptability of robots [43], [44]. Empathetic agents have also even been shown to increase the chances that a person donates to charity [45].

In social robotics, empathy has been explored as a characteristics for robots to increase their acceptance and educational performance in various scenarios [46], [47], [48]. Likewise, social robots such as Haru have been studied specifically on empathetic abilities [49], [50].

In this review, we focus on the most basic application of the field: open-domain conversations. We believe that this is an application that predates specialization and is much more flexible and representative of the concept of empathy in itself.

V. DATASETS

We present the datasets used for empathetic text generation (ETG) in this section. A previous survey has explored some of these and their uses in specific algorithms [12]. Therefore, we focus on datasets not included in preceding surveys.

In addition, we present caveats for using the currently most popular ones.

The most popular dataset for ETG is EMPATHETIC-DIALOGUES [13], a dataset of around 25k open-domain one-on-one conversations. It is centered on 32 emotion labels with prompts related to them. It is widely used today for training and testing algorithms [51], [52]. However, other datasets have been proposed to address its shortcomings, particularly its length [12], [53]. Some include empathy cues or increase their size, such as Persona-based Empathetic Conversations (PEC) [54], Emotional Dialogues in OpenSubtitles (EDOS) [55], XiaoAI Empathetic Conversation (X-EMAC) [56], and Empathetic OpenSubtitles [57]. BlendedSkillTalk [58] uses its data alongside other databases to make a multiskill conversational agent.

EMPATHETICDIALOGUES, and its inspired successors [54], [55], [56], [57], are useful for ETG. However, it has unaddressed downsides that show that new datasets are still needed. For example, some algorithms have been shown to generate repetitive responses [59]. A critical analysis by Debnath [60] showed questionable validity for some of the 32 emotion labels in the dataset. Similarly, there is limited evidence to prove the correlation between the label of a conversation, its prompt, and an utterance. Finally, there is no stated method for validating this corpus. Therefore, there is no way to address whether empathy is captured well. Montiel-Vazquez et al. [61] substantiate this point, as their research showed low empathy ratings in many conversations. Automatic metrics of empathy are being developed currently [9], [11], [32], [62], [63], [64]. Therefore, we expect future research to address whether the database captures empathy.

Novel datasets present different methodologies for obtaining data. An important one to address is using Large Language Models (LLMs). LLMs are a cutting-edge paradigm for text generation [18], [65]. They are based on Transformers, a neural network architecture used for sequence processing [66]. LLMs use large databases for predicting text sequences [65]. LLMs have performed remarkably in various Natural language processing (NLP) tasks [51]. As such, they can be leveraged to build novel databases for this task.

Zhang et al. [18] constructed a database using a multi-agent LLM system called Agent for StickerConv (Agent4SC). The database, StickerConv, contains around 13k dialogue sessions that also leverage stickers as a communication tool for increasing empathy. The advantages of this database are that it acknowledges stickers as a tool for enriching emotion, is built without manual annotation, and addresses multi-personality variables. However, it is still relatively small even when compared to EMPATHETICDIALOGUES, which is small compared to databases used for LLMs.

Another relevant database that uses LLMs is AUGESC, or **Aug**mented **E**motional **S**upport **C**onversations [67]. It was created by augmenting the ESConv corpus [68], which contains conversations giving emotional support. Emotional support conversations can include elements of emotional chatting, and empathetic responding. The quality control and need for worker training for ESConv led to it containing only 1.3K dialogue sessions. To alleviate data scarcity in the dataset, Zheng et al. use an LLM to complete dialogues found in EMPATHETICDIALOGUES with negative emotion labels. Their approach successfully augmented AUGESC to be 45 times larger than ESCov and to include more topics. This method was evaluated using a comprehensive human judgment. The success of AUGESC shows the feasibility of using LLM data augmentation for subjects related to empathy.

An approach for ETG that is seldom explored is multi-party conversations. As such, few datasets are available. Zhu et al. [69] crafted a Multi-party empathetic dialogue (MPED) database from an online peer-to-peer support platform. The most relevant contribution is that MPED contains conversations with multiple participants, instead of being limited to one-on-one engagements. This database already contained emotion labels and was further processed by the authors to include the emotional reaction, interpretation, and exploration levels of responses [32]. Besides, MPED includes single-turn and multi-turn dialogue data, called MPED-S and MPED-M.

Finally, it is worth mentioning the databases that have not been used extensively for ETG, but have potential. For example, those that are used to measure the levels of empathy in text. Some notable ones are the datasets based on EPITOME [32], EmpatheticConversations (EC) [61], and Empathetic Conversations [64]. Using these datasets directly for ETG or to evaluate the results from ETG algorithms could yield insight into this field. Additionally, Debnath [60] presents the *incidental empathy hypothesis* that postulates that datasets with human interactions, like DailyDialog [70] and PERSONA-CHAT [71], must contain cases of empathetic engagements that we could identify and isolate for ETG.

There are many different datasets for ETG. Each contains its unique features and downsides. With the development of Affective computing as a field, we can expect future research to expand on the available datasets. The advantages of tools like LLMs, the availability of features, and the diversity of approaches show great potential for improving and creating new datasets for ETG. Leveraging the available datasets with their respective features and using LLMs to enhance the data could be an approach that yields a new dataset, like EMPATHETICDIALOGUES, without many shortcomings.

VI. SOURCING METHODS FOR EMPATHY CUES IN ETG

In this section, we explore the methodologies to obtain empathy cues as numerical, categorical, text, or vector variables to use in ETG systems [9], [12]. The task presented in this section consists of obtaining these variables from a text utterance. For example, applying a classification algorithm to obtain the emotion expressed, or a language model to extract semantic relations.

Empathy cues can be any concept that contributes to behavioral empathy. Therefore, these methods are diverse. In our survey, we identify four methods: (A) using humans to perform *manual annotation* of empathy cues, (B) *retrieving* the values of empathy cues from external databases, (C) applying *classification* algorithms to get the values for empathy cues, and (D) using *generative models* to extract the values of empathy cues from a text input. These methods are used in the task of extracting or inferring features from text utterances. Therefore, these **sourcing methods** are considered external and supplementary applications of machine learning, datasets, and human judgment to the NLP pipeline, as observed in Figure 1. They are not interchangeable, as each method provides distinct types of cues. We explore them individually in the following subsections.

A. MANUAL ANNOTATION

This sourcing method consists of using human judgment to manually annotate the values of empathy cues. Innovative affective computing research sometimes requires new data [13], [21], [53]. Therefore, researchers rely on human annotators to obtain it. Manual annotation is often performed during the creation of a dataset. However, it can be required in the text generation process. We present a summary of the empathy cues and models that use manual annotation to source empathy cues in Table 1.

When building a new dataset, researchers will allow humans to create data points following an empathy cue of interest. This is how many relevant datasets were obtained, including ED [13], Japanese EmpatheticDialogues (JPED) [72], and the Reddit Empathy-Mental-Health dataset [32]. During dataset creation, the most likely cue that must be manually annotated is emotion. However, specific datasets and models are also built on other features that were obtained by human judges [53].

In the case of a dataset that is already available but does not have the cue of interest, researchers have relied on manually labeling instances belonging to that dataset, such as ED [13]. This case is observed with emotion-cause (the part of the sentence that causes the overall emotion) [31], [56], [73]. Likewise, a dataset can be designed to complement another. For example, Fu et al. [74] manually create responses for the 32 emotion categories in JPED [72].

If manual annotation for an entire dataset is impossible, researchers can rely on partial annotation. In it, human annotators will assign values for empathy cues to a subset of a database. Subsequently, an algorithm will be trained to label all instances based on those values. This was the case of one of the most common cues in our survey: empathetic intent, the intent of an empathetic speaker towards another person with their communication [55]. Welivita et al. [53] partially annotate ED instances with empathetic intent. Then, label the rest of the dataset using a classification algorithm. Similarly, Pang et al. [73] mark whether a response aims to validate a speaker in the JPED dataset.

Finally, some approaches use manually annotated variables in their generation processes. The emotional chatting machine (EMC) [21], required manual selection of an emotion to guide text generation. Likewise, cues can be appended to the input of ETG, like a persona, emotion, or action labels [30], [75]. Nevertheless, that manual selection of features is impractical for new data points [15]. Therefore, this is rarely observed in the literature.

The most common cues obtained through manual annotation for ETG systems are emotion and emotion-cause. Nevertheless, some require data that is much more subjective or that requires higher scrutiny. For example, validation timing, personality, and example responses are proposed by these ETG algorithms and rely on the judgment of humans to increase their empathetic performance. In the case of Fu et al. [74], manual annotation was required to supplement the generation process with pre-written responses. This ensures more ''human'' dialogue responses. However, it makes the algorithm more constrained. Dynamical adaptation of these algorithms is difficult, given that these models work under the constraints of the labels provided. This strategy for empathy cue sourcing is less popular than others. Despite that, the data produced by these algorithms is used as training data in other ETG approaches.

Manual annotation is still a valuable method for obtaining empathy components as features. The field of affective computing is still developing robust definitions [9]. Furthermore, empathy and empathy cues such as emotions, whether a person requires validation, and persona can still be considered subjective and, in some cases, vague [76]. As such, human judgment of these methods might ensure accuracy. Nevertheless, it is time-consuming and sometimes requires laborious worker training and human screening [67]. An alternative approach for manual annotation could be found in LLMs, which can be used to build and augment datasets with comparable results [18], [67].

B. RETRIEVAL

We call retrieval the method of sourcing empathy cue values from external datasets. These can be datasets generated for the specific model of that have been repurposed for ETG. An important characteristic of this method is that the external dataset is not used to train or test the model for empathetic text generation. These datasets are merely sources for values for empathy cues of interest. Some examples are emotion lexicons [78], [79], [80] and commonsense databases [79], [80], [81]. In these cases, cues from external datasets are used to enrich the knowledge of the core ETG algorithm and leverage it to respond more empathically. We present a summary of the models, according to the external dataset used in Table 2.

The first approach we observed to use this method was EmpDG [78]. This algorithm leverages knowledge from a single database: NRC Emotion Lexicon (NRC) [82]. Li et al. extract words considered ''emotional'' and, alongside

TABLE 1. Overview of models that use manual annotation in their text generation process. Only best performance using automatic metrics is reported in Results. PPL: Perplexity. DS-n: Distinct-n. F: F1 score.

Model	Cues	Metric	Results	Training Data
EMC [21]	Emotion	6 Labels	PPL: 61.8	NLPCC dataset Emotional STC Dataset [21]
EmpTransfo [30]	Emotion Action Topic	7 Labels 4 Labels 10 Labels	PPL: 78.47, Hit@1: 78.47, BLEU: 2.45, F: 17.27	DailyDialog [70]
CAiRE [75]	Emotion Persona	Labels Custom Sentences	PPL: 13.32, BLEU: 7.03,	ED [13] PERSONA-CHAT [71]
EMMA [56]	Emotion	Labels	PPL: 16.99,	ED [13]
Validating responses [73]	Emotion Emotion-cause Validation timing	8 Labels Text Binary Label	BERTscore_JEPD: 59.34 BERTscore_TESC: 57.18	JPED [72] TESC [77]
ECTG [31]	Emotion-cause	Predicted concepts	BLEU-4: 5.467 BERTscore F: 0.2701 DS-1/2: 1.840/16.404 ROUGE-L: 23.77 CIDEr: 51.43	ED [13]
Fu et al. [74]	Examples	Pre-written responses	BLEU-4: 5.467 DS-1/2: 2.20/7.99	JPED [72]

adjectives not in the lexicon, use them as a feature of EmpDG. With this, Li et al. obtain fine-grained token-level emotions that can be used alongside the coarse-grained emotion labels in ED [13].

Beyond, EmpDG, more recent algorithms leverage multiple databases for ETG [79], [80], [83]. The most common databases used by this algorithm family are the commonsense knowledge graph ConceptNet [84], and the emotional lexicon NRC_VAD [85].

ConceptNet [84] is mainly used to extract keywords and it can be combined with another dataset to obtain semantic and emotional knowledge. Liu et al. [79] pioneered the use of it and NRC_VAD [82]. In their work, keyword-concept tuples are obtained based on emotional intensity, semantic similarity, and a scaled confidence score. Then, these tuples are input to a Transformer decoder as plain text. The use of these datasets for retrieval is also observed in algorithms such as KEMP [80], EMO_SA [83], and CAB [52]. However, these use keyword and emotion tuples to predict emotional control signals, compute emotion frequency, and enrich their context graphs. In general, ConceptNet alongside a source of emotional words can imbue ETG algorithms with contextual and affective knowledge based on specific words.

Besides ConceptNet and emotion lexicons, other databases allow algorithms to obtain empathic cues. For example, semantic frames are obtained by Bi et al. [17] using the FrameNet linguistic knowledge graph [86], [87]. Meanwhile, text responses and exemplars from datasets built from ED and specific datasets are used by various algorithms to guide the ETG process and ensure consistent responses [17], [51], [74], [76]. An example of the latter approach using LLMs is proposed by Sotolar et al. [88], which uses examples and counter examples based on opposite emotions to provide each prompt with preferred and non-preferred completions.

The cues extracted using the retrieval method are overwhelmingly related to the semantic and emotional characteristics of the text. This type of knowledge is accessible due to the nature of the databases used in the surveyed ETG approaches. They mostly re-purpose available lexical tools. ConceptNet and the NRC_VAD lexicons are useful in conjunction for many of these algorithms. Nevertheless, they can also be used separately. The main advantage of using a retrieval strategy is that there is no necessity for carrying out manual annotation or verifying the accuracy of your sources of knowledge. Furthermore, these datasets are readily accessible for integration into an ETG algorithm. However, there are also disadvantages such as the need for constructing complex graphs using the datasets, large retrieval times, and problems like unreliability of the datasets.

Another trend we observe in these cues is the retrieval of exemplars and similar situations. In this case, the approaches mirror the methodology taken by the human brain: Similar responses and examples are obtained given a criterion and they are subsequently applied to dialogue generation. As such, this methodology tends to work well within the same contexts as those in their databases. Nevertheless, we observe the possibility of repetitive responses.

Retrieval is a method that aims to leverage existing knowledge repositories to enrich dialogue generation models [80]. As such, we observe a trend in which models that use this strategy aim to extract finer details and nuances of a conversation [83] using external datasets. In general, this methodology aims to replicate the process using past experiences and memories for formulating more empathetic responses.

C. CLASSIFICATION

This subsection presents the ETG systems that use classification algorithms to obtain the labels of empathy cues.

TABLE 2. Overview of models that use retrieval in their text generation process. Only best performance using automatic metrics is reported in Results. PPL: Perplexity, AC: Accuracy, F: F1 score, DS-n: Distinct-n, EMO: Emotion, I: Intent, ER: Emotional Reaction, IP: Interpretation, EX: Exploration, NIDF: Normalized Inverse Document Frequency.

Model	Cues source	Cues	Metric	Results	Training Data
EmpDG [89]	NRC Emotion Lexicon [82]	Emotion Words	Tokens, Labels	AC: 0.3431 PPL: 34.18 DS-1/2: 1.81/6.94	ED [13]
RGPT2 + CKECE [79]	ConceptNet [84], NRC_VAD [85]	Emotion Words	Keyword-concept Tuple	AC: 0.5151, PPL: 13.57 DS-1/2: 2.04/11.68	ED [13]
KEMP [80]	ConceptNet [84], NRC_VAD [85]	Emotion Knowledge	Emotion Graph	AC: 0.3931 PPL: 36.89 DS-1/2: 0.55/2.29	ED [13]
EMO_SA [83]	ConceptNet [84], NRC_VAD [85], ED [13]	Emotion Knowledge Emotion Tendency	Emotion Graph Emotion correlation	AC: 0.402 PPL: 34.51 DS-1: 0.78/24.58	ED [13]
CASE [90]	ConceptNet [84], NRC_VAD [85]	Emotional concepts	Graph	PPL: 35.37 DS-1/2: 0.77/4.01	ED [13]
CAB [52]	ConceptNet [84]	Semantic knowledge	Keyword Entities Keyword Relations	PPL: 34.36 DS-1/2: 1.13/4.23	ED [13]
DiffusEmp [17]	FrameNet [86], [87], ED [13]	Semantic Knowledge Candidate responses	Semantic frame Text	BERTscore: 0.5101 MIscore: 626.92 DS-1/2: 1.79/26.97 DS-4: 88.17, F_BLEU: 1.29	ED [13] EmpatheticIntents [53] EMH [32]
Fu et al.'s model [74]	Model dataset [74]	Examples	Pre-written responses	BLEU-4: 5.467 DS-1/2: 2.20/7.99	JPED [72]
LEMPEx [76]	ED [13]	Examples	Responses	BLEU: 7.88, PPL: 26.37 DS-1: 1.41, DS-2: 14.66	EMH [32] ED [13]
EmPO [88]	ED [13]	Examples Counter Examples	Text Text	BLEU: 7.88, PPL: 26.37 DS-1: 1.41, DS-2: 14.66	ED [13]
EDG_MultiEmoInt [91]	NRC_VAD [85]	Valence Arousal Dominance	[-1,1] [-1,1] [-1,1]	PPL: 36.88 DS-1/2: 0.51/2.12 AC: 37.41	ED [13]
EmpGTP-3 [51]	ED [13]	Situations with similar emotion	Text	PPL: 118.86, DS-1/2: 0.94/0.99 EMO/I_AC: 0.25/0.15 NIDF: 0.28, IP: 0.27, EX: 0.49 ER: 0.64	ED [13]

These are a type of machine learning algorithm used to get a label, value, or judgment from given data. They separate a dataset into different categories [92]. We see classification algorithms integrated into the ETG pipeline extensively. They are found in the input [13], core text generation algorithm [14], and output [76] layers. Using classification for empathy cues predates EMPATHETICDIALOGUES (ED) [13], as it is seen in antecedents such as XiaoIce [20], a large-scale chatbot which uses them to obtain user information. We present a comprehensive exploration of all ETG models that incorporate classification in Table 3.

The model that popularized classifiers in ETG was Emo-Prepend by Rashkin et al. [13]. This model uses a classifier to obtain 1 of 32 emotion labels from a text input. This strategy would be adopted by various models [14], [15], [16], [24], [52], [73], [74], [90], [93], [94], [95], [96], [97], [98], [99]. Using emotion classifiers allows models to obtain an emotional label from a text utterance, allowing them to acknowledge and act according to a user's emotion. Empathy is heavily related to emotions through its affective dimension. Therefore, emotion labels are still obtained in the latest ETG methods [100].

Sentiment is another variable related to the affective dimension of empathy [61]. As such, we see sentiment labels obtained through classifiers in ETG [14], [101]. This can be done through mapping emotions in ED [76], [101] or using a pre-trained classification algorithm [76], [102]. Sentiment classifiers are limited to polarity labels that do not exceed three values: positive, neutral, and negative.

An empathy cue obtained almost exclusively through classification is empathetic intent [16], [57]. This cue consists of the intent of an empathetic response, usually based on a taxonomy of intents first proposed by Welivita and Pu [53]. This has mostly been achieved through a transformer-based classification algorithm [103]. Many ETG proposals use this classifier to obtain a label out of 8 intents plus a neutral [16], [17], [52], or 8 intents and a neutral plus 32 emotions [57], [103], [104]. This method is often used to guide text generation, as empathetic intent is associated with the behavioral dimension of empathy [52]. A related approach is obtaining labels for dialogue strategy [105], which was used by Liu et al. [81].

The EPITOME framework [32] is one of the most important developments in empathetic classification and ETG. It was developed by Sharma et al. and it is composed of three models that predict empathetic communication mechanisms, and their rationales, in an interaction: (1) Emotional reaction, (2) Interpretation, and (3) Exploration. (1) Emotional reaction refers to a user expressing emotions, (2) Interpretation refers to the act of stating understanding of feelings and the experiences of another, and (3) Exploration refers to communicating questions and showing interest. This framework is adopted by various algorithms in the literature, with each mechanism used as individual empathy cues [17], [96], [104], evaluation metrics [106], and loss functions [76].

TABLE 3. Overview of models that use classification in their text generation process. Only best performance using automatic metrics is reported in Results PPL: Perplexity. SES: Sentence embedding similarity. AC: Accuracy. P: Precision. R: Recall. F: F1 score. DS-n: Distinct-n. L: Listener. S: Speaker. EMO: Emotion. I: Intent. CM: EPITOME communication mechanisms. EAD: Expectation Adjusted Difference. EmbEx: Vector extrema cosine similarity.

Model	Cues	Metric	Results	Training data
EmoPrepend [13]	Emotion	32 Labels	PPL: 24.30 AVG BLEU: 4.36	ED [13]
MoEL [15]	Emotion	32 Labels	BLEU: 2.90	ED [13]
EmoAwareTrs [93], [107]	Emotion	32 Labels	BLEU: 0.225	ED [13]
CEM [95]	Emotion	32 Labels	PPL: 36.11 DS-1/2: 0.66/2.99	ED [13]
Fu et al. [74]	Emotion	32 Labels	BLEU: 6.23 DS-1/2: 2.20/7.99	JPED [72]
CASE [90]	Emotion	32 Labels	PPL: 35.37, AC: 40.2 DS-1/2: 0.77/4.01	ED [13]
EmpSOA [108]	Emotion	32 Labels	PPL: 35.02, AC:48.32 DS-1/2: 0.71/3.96	ED [13]
EmpSOI [99]	Emotion	32 Labels	PPL: 34.79 DS-1/2: 0.79/4.24, AC: 41.52 $P/R/F_{BERT}$: 0.86/0.85/0.85	ED [13]
EmpCRL [109]	Emotion	7 Labels	PPL: 15.7, AC:41.57 DS-1/2: 4.27/16.11 EAD-1/2: 5.39/22.63	ED [13]
PECER [98]	Emotion	7 Labels	PPL: 14.42, DS-1/2: 3.52/16.95 BLEU-1/2/3/4: 21.12, 10.01,5.52,3.40	ED [13]
Sentiment Look-Ahead [101]	Sentiment	Binary label	BLEU: 6.317 DS-1: 0.026, DS-2: 0.170 DS-3: 0.379	JPED [72] DailyDialog [70] PERSONA-CHAT [71]
CKEDG [81]	Dialogue strategy	7 Labels	BLEU-1/2: 20.12/8.2 BLEU-3/4: 4.16/2.47 BERTscore: 90.97, AC: 16.29	ESConv [68]
TopicPrepend [13]	Topics	20 Labels	PPL: 25.40 AVG BLEU: 4.17	ED [13] 20-Newsgroup dataset [110]
MIME [14]	Emotion Sentiment	32 Labels Binary Label	BLEU: 2.98	ED [13]
RQL [100]	Emotion Sentiment	6 Labels Binary label	PPL: 18.01, BLEU-2/4: 7.20/3.20, DS-1/2: 0.0106/0.1695,	EMC [21]
MEED [57]	Emotion Intent	32 Labels 9 Labels	PPL: 28.5, SES: 0.171 DS-1/2: 0.070/0.255 P: 0.10,R: 0.14,F: 0.09	ED [13] EDOS [55] OpenSubtitles [111]
EmpHi [16]	Emotion Intent	32 Labels 9 Labels	F_BLEU: 0.382 DS-1/2: 1.11/5.33	ED [13] EmpatheticIntents [53]
SEEK [24]	Emotion Intent	32 Labels 9 Labels	EMO_AC: 41.85, I_AC: 34.08 PPL: 37.09, DS-1/2: 0.73/3.23	ED [13] EmpatheticIntents [53]
ConIntrEmp [103]	Emotion Intent	32 Labels 9 Labels	PPL:13.23, EmbEx:0.48 DS-1/2: 0.08/0.29	ED [13] EDOS [55]
CAB [52]	Emotion Intent	32 Labels 8 Labels	PPL: 34.36 DS-1/2: 1.13/4.23 EMO_AC(S/L): 40.52/72.23 I_AC: 41.72	ED [13] EmpatheticIntents [53]
EmpCI [112]	Emotion Intent	32 Labels 9 Labels	PPL: 35.08, AC:40.86 DS-1/2: 0.83/4.38	ED [13] EmpatheticIntents [53]
Gao et al. [94]	Emotion Emotion-cause	32 Labels Binary Label	BLEU: 1.774, AC: 42.4 DS-1/2: 0.017/0.084 $P/R/F_{BERT}$: 0.145/0.127/0.136	ED [13] RECCON [113]
EMP-RFT [96]	Emotion EPITOME	32 Labels Triple 3-level Scales	PPL: 13.59, F_{BERT}:0.34 DS-1/2: 3.24/4.59	ED [13] EMH [32]
RLCA [114]	Emotion EPITOME	7 Labels Triple 3-level Scales	PPL: 31.1, EMO_AC: 68.45 DS-1/2: 0.70/3.11 $P/R/F_{BERT}$: 0.86/0.85/0.85	ED [13], CC [115] DailyDialog [70] PERSONA-CHAT [71]
LEMPEx [76]	EPITOME Sentiment	Triple 3-level Scales [-1,1]	BLEU: 7.88, PPL: 26.37 DS-1: 1.41, DS-2: 14.66	EMH [32] ED [13]
Validating responses [73]	Emotion Emotion-cause, Validation timing	8 Labels Text Binary Label	BERTscore_JEPD: 59.34 BERTscore_TESC: 57.18 EMO_AC: 77.2	JPED [72] TESC [77]
DiffusEmp [17]	Emotion Intent EPITOME	32 Labels 9 Labels Triple 3-level Scales	BERTscore: 0.5101 MIscore: 626.92 DS-1/2/4: 1.79/26.97/88.17 F_BLEU: 1.29, CM/I_AC: 92.36/84.24	ED [13] EmpatheticIntents [53] EMH [32]
EPIMEED, EPIMEED+ [104]	Emotion Intent EPITOME	32 Labels 9 Labels Triple 3-level scales	BLEU-1: 0.054, BLEU-2: 0.0191 ROUGE-L: 0.0559 METEOR: 0.063 DS-1/2: 0.0039/0.0018	ED [13] EmpatheticIntents [53] EMH [32] RED [104]

Classification algorithms have been used to supplement, guide, and evaluate empathetic text generation. To do this, they predict a label from a series of options. Subsequently, this prediction is incorporated into an ETG algorithm to improve their empathic performance. The nature of these cues are of elements directly related to empathy dimensions that have been previously categorized. As such, we observe newer algorithms re-purpose previously used classifiers into their text generation pipeline.

It is important to note that classification requires manually annotated categories, which vary between training datasets. As such, similar cues are not necessarily equivalent. For example, we observe different implementations for empathetic intent [53], [105]. Furthermore, the reliability of a cue is dependent on the labels and the classification algorithm used. Therefore, there is discrepancy between the implementation of a cue. For instance, emotion classifiers can vary in the number of predicted labels.

An advantage of classification algorithms is that new data can be labeled with an empathy cue automatically. As such, they can be used with inputs different than those of their training database. Therefore, they are a versatile method to construct databases. Furthermore, the categories provided by these classification algorithms are based on psychological research [13], [32], [53]. Thanks to their advantages and ease of implementation, they are the most popular method for introducing empathy cues to ETG approaches. Nevertheless, it is important to note that classifiers are mostly limited to the cues explored in this section. The process of creating a novel classifier requires developing a taxonomy for the cue of interest, gathering data for a dataset, and training the algorithm. As such, while this methodology is widespread, there is not much variety in classifiers used for empathy cues.

D. GENERATIVE MODELS

Extracting more complex empathy cues for ETG has required researchers to leverage generative models in their algorithms. This family of models contains Transformers [66], [116], Diffusion Models [17], Large Language Models (LLMs) [18], [65], and others [117], [118]. We identified that transformer-based models are frequently used to obtain values for empathy cues. However, diffusion approaches have also been successful. Generative models are a much more broad category of algorithms. Therefore, the nature of the cues obtained through this strategy is more diverse. Nevertheless, we observe some trends in their application. We present the ETG approaches, their supplementary generative models, and their empathy cues in Table 4.

One of the most popular applications of generative models is to obtain emotion-cause keywords [31], [94], [96], [106]. Kim et al. [106] created a generative emotion estimator that uses a transformer-based model to extract keywords using Bayes' theorem [119] from the utterances of a person on the *speaker* role of a conversation. This method outperformed an automatic keyword extractor [120] and the BERT [121] attention weights. Subsequently, this same model was later adopted by Kim et al. [96] for the Emp-RFT algorithm, which also used a RoBERTa-based model [32] to extract keywords and emotions from the *listener* role. Then, Kim et al. propose combining both sets of keywords to create keyword pairs. Likewise, Gao et al. [94] use a sequence labeling model for detecting emotion-cause keywords. Meanwhile, Qian et al. [31] separate keywords in the emotion cause to create a graph representing transitions between keywords.

Another cue that is highly sought-after in ETG is commonsense knowledge. This has been proposed to adopt cognitive empathy into the ETG pipeline [81]. The most common method for extracting commonsense relations is the COMET language model [116]. Particularly, trained on the ATOMIC dataset [122]. This model obtains commonsense relations (e.g xIntent, xNeed, xWant, xEffect) which are them used as inputs or to construct graphs. This method allows the algorithm to be aware of commonsense knowledge during response generation. The first case, to our knowledge, was proposed by Sabour et al, with the CEM algorithm [95]. Nevertheless, this same methodology has been adopted by many other posterior models [24], [90], [99], [108], [112].

Another generative method that can be used to address empathy cues is LLMs. Zhang et al. [18] present an approach based on the Vicuna LLM [123] called PEGS (Large multimodal dialog model with the ability to **PE**rceive and **G**enerate **S**tickers). PEGS first encodes the images [124], passes them through Query Transformer [125] developed for LLMs, and then uses a linear layer to obtain a representation that can be sent to the Vicuna LLM. Likewise, approaches like those in [98] and [109] obtain commonsense cause sentences and emotional contexts using a generative model and the CICERO dataset [126]. In the case of PECER, a fine-tuned LLM can be used to obtain personality descriptions from the dialogue history.

Finally, we identified two approaches that use linguistic-based machine learning models to improve the performance of ETG. The first, by Yang et al. [117] uses a dependency parser [127] to build an internal dependency tree to incorporate the relations of emotion and semantics in text generation. Meanwhile, Li et al. [118] use a pre-trained model obtained on the STAC dataset [128] to obtain discourse characteristics according to Segmented Discourse Representation Theory (SDRT).

The method of using generative models leverages the advantages of SOTA developments [32], [66], [121] for labeling parts of the sentence and obtaining more knowledge. Obtaining knowledge in this way allows the core text generation algorithm to improve the behavioral empathy of the model with innovative cues. It is an alternative to using databases to include knowledge in the text generation pipeline. The main advantage of this strategy is that these models can obtain relevant knowledge quickly

TABLE 4. Overview of models that use generative models in their text generation process. Only best performance using automatic metrics is reported in Results PPL: Perplexity. SES: Sentence embedding similarity. DS-n: Distinct-n. RED: Reddit Emotional Distress. CVR: Coverage of keywords in response. AC: Accuracy. P: Precision. R: Recall. F: F1 score.

Model	Supplementary Model	Cues	Metric	Results	Training Data
EC framework [106]	BART [129]	Emotion-cause	Keywords	CVR: 0.54 EX: 0.38 IP: 0.26	ED [13] DodecaDialogue [130] BlendedSkillTalk [58]
Gao et al. [94]	RoBERTa [131]	Emotion-cause	Sentence labeling	BLEU: 1.774,AC: 42.4 DS-1/2: 0.017/0.084 $P/R/F_{BERT}$: 0.145/0.127/0.136	ED [13] RECCON [113]
EMP-RFT [96]	BART [129] RoBERTa [32]	Emotion-cause Emotion-cause	Keywords Keywords Keyword-pairs	PPL: 13.59, F_{BERT}:0.34 DS-2: 4.59	ED [13] EMH [32]
ECTG [31]	SpanBERT [132]	Emotion-cause	Sentence labeling	BLEU-4: 5.467 BERTscore F: 0.2701 DS-1/2: 1.840/16.404 ROUGE-L: 23.77 CIDEr: 51.43	ED [13]
EDG_MultiEmoInt [91]	COMET [116]	Emotion Intent	Concept relations Concept relations	PPL: 36.88, AC:37.41 DS-1/2: 0.51/2.12	ED [13]
CKEDG [81]	COMET [116]	Commonsense knowledge	Text Prompts	BLEU-1: 20.12, BLEU-2: 8.2 BLEU-3: 4.16, BLEU-4: 2.47 BERTscore: 90.97	ESConv [68]
CEM [95]	COMET [116]	Commonsense knowledge	Concept relations	PPL: 36.11 DS-1/2: 0.66/2.99	ED [13]
SEEK [24]	COMET [116]	Commonsense knowledge	Concept relations	PPL: 37.09 DS-1/2: 0.73/3.23	ED [13] EmpatheticIntents [53]
CASE [90]	COMET [116]	Commonsense knowledge	Concept relations	PPL: 35.37 DS-1/2: 0.77/4.01	ED [13]
EmpCI [112]	COMET [116]	Commonsense knowledge	Concept relations	PPL: 35.08, AC:40.86 DS-1/2: 0.83/4.38	ED [13] EmpatheticIntents [53]
RLCA [114]	COMET [116]	Commonsense knowledge	Concept relations	PPL: 31.1 DS-1/2: 0.70/3.11	ED [13], CC [115] DailyDialog [70] PERSONA-CHAT [71]
EmpSOA [108]	COMET [116]	Commonsense knowledge	Concept relations	PPL: 35.02, AC:48.32 DS-1/2: 0.71/3.96	ED [13]
EmpSOI [99]	COMET [116]	Commonsense knowledge	Concept relations	PPL: 34.79, F_{BERT}:0.85 DS-1/2: 0.79/4.24 P/R_{BERT}: 0.86/0.85	ED [13]
EmpCRL [109]	ChatGPT-CICERO [126]	Commonsense knowledge	Cause sentences	PPL: 15.7, AC:41.57 DS-1/2: 4.27/16.11 EAD-1/2: 5.39/22.63	ED [13]
Model + Sibyl [133]	Sibyl [133]	Commonsense knowledge	Cause Subsequent Event Emotion State Intention	BLEU-1/2/3/4: 21.4/9.3/5.01/2.95 ROUGE-L: 19.08, METEOR: 9.61 DS-1/2/3: 5.65/36.11/72.02 CIDEr: 26.93	ED [13] ESConv [68]
PECER [98]	DIALeCT -CICERO [126] T5 [134]	Commonsense knowledge Personality	Emotion context sentences Text description	PPL: 14.42, DS-1/2: 3.52/16.95 BLEU-1/2: 21.12/10.01 BLEU-3/4:5.52/3.40	ED [13] CICERO [126] PERSONA-CHAT [71]
PEGS [18]	Vicuna LLM [123] Stable Diffusion [135] ViT-G/14 [124] QFormer [125]	Communication strategy	Stickers	BLEU-1/2/3/4: 0.47/0.33/0.26/0.21 DS-1/2/3: 0.84/0.99/0.99 ROUGE-L: 0.35, METEOR: 0.44 CIDER: 0.57, BERTScore: 0.911	StickerConv [18]
ESCM [117]	Biaffine Parser [127]	Emotion-Semantic correlations	Graph	BLEU-4: 5.467 PPL: 34.82 DS-1/2: 1.19/4.11	ED [13]
InfRa [118]	SDT Parser [136]	SDRT relations correlations	Graph	BLEU: 0.58, MIScore:3609.25 PPL: 25.87, BERTScore: 58.89 DS-1/2/4: 3.7/22.41/62.61	ED [13]

and accurately. However, applying generative algorithms might increase the computational complexity of the model exponentially.

VII. INTEGRATION STRATEGIES

This section describes the trends and techniques to incorporate empathy cues in text generation applications. We first describe the most straightforward method. Then, methods that use a variable as a condition for guiding ETG. Finally, we explore hybrid strategies. This order can be seen as a development timeline. However, many of these methods are contemporaneous. We present a summary of the models explored by our survey in Table 5.

A. CONTEXT AUGMENTATION

The first category of integration strategies is context augmentation. It consists of concatenating empathy cues to the input of the system, allowing it to learn the relationship between

these features and the text. This is expected to increase the perceived empathy of responses [13], [52], [75]. This method is straightforward and predates EMPATHETICDIALOGUES (ED) [20]. However, it is still being used to ensure that the system acknowledges certain empathy cues [51], [100].

The most common empathy cue incorporated through this strategy is emotion. A model can concatenate an emotion label alongside the text input, as observed with EmoPrepend [13]. This has been achieved by performing emotion classification alongside text generation [75], [30], [75], [95]. In this manner, the model is made aware of the *speaker* emotion, as well as the dialogue history. Emotion is almost universally considered in the context augmentation strategy for ETG and it can be observed from the earliest [93] to the newest models [99].

Context augmentation has also been observed with other types of empathy cues. For example, a persona for both the speaker and the listener can also be concatenated to the dialogue history input [98]. In the CAiRE model, a manually created persona, alongside emotions, is input to the dialogue history [75]. This ensures that the model has a consistent personality. However, it makes it so that vital information for the model must come from manual annotation.

Cues sourced from generative algorithms are also integrated through this strategy. For example, CEM [95], SEEK [24], EDG_MultiEmoInt [91], and EmpSOA [108] use pre-trained language models to integrate commonsense relations into the generation process, alongside emotion from classification or lexicons. An interesting aspect of EmpSOA is that it is one of the few to separate the attributes from the self and the other, which grants it self-other awareness.

Finally, LLMs also allow for integration of complex knowledge. Particularly, text can be concatenated to an input prompt to enrich dialogue context. Examples of this strategy can be seen in EmpGPT-3 [51], and PECER [98]. These algorithms introduce similar emotional situations, and emotional context sentences with personality descriptions into their inputs. In this way, the authors make the algorithms aware of these variables, which improves their performance.

Context augmentation was the first methodology to incorporate empathy cues into the text generation task. As such, it is straightforward and does not require majorly modifying the architecture of a text generation algorithm. It is easy to implement. However, this does not mean that it is not effective. We observe that this methodology is particularly effective to handle complex and new empathy cues into the ETG pipeline.

Many SOTA algorithms still rely exclusively on this integration strategy [51], [98]. Nevertheless, we observe that it is also used to complement other cues in models with more complex structures, or that aim to cover a wider set of empathy cues. For example, CAB [52] and DiffusEmp [17].

B. VARIABLE CONDITIONING

We define variable conditioning as a cue integration strategy that uses variables to explicitly control the generation process. Most of the ETG algorithms in this survey incorporate this strategy to ensure that the response is dependent on one or multiple empathy cues. Therefore, the empathy cue directly influences the generation strategy [52]. The contrast between context augmentation and variable conditioning in small (transformer-based) language models lies in the modification of the model architecture to influence the response, rather than just including the cue values in the input. In the case of LLMs, variable conditioning occurs when the prompt is explicitly told to incorporate the cue into the response.

Variable conditioning has antecedents in other applications, such as emotional dialogue. For example, The Emotional Chatting Machine (EMC) used manually specified emotion labels to condition the creation of text [21]. This strategy has since been adopted and expanded in ETG [103]. Currently, SOTA algorithms use automatically predicted sentiment labels [14], [101], emotion [15], intent [16], [103], and other empathy cues as signals for controlling ETG.

Methods that condition on emotion labels were among the first to try this approach. MoEL [15] uses emotion classification, alongside multiple listeners trained on each emotion, to obtain a balanced empathetic response. Whereas MIME [14] uses emotion classification results and the polarity of the emotion to select when to mimic an emotion or not.

Reinforcement learning is an approach that ensures responses align with one or more empathy cues. In this type of machine learning, agents learn from interacting with their environment; a reward function defines the goals of the agent [137]. For ETG, a cue is used as a reward function. For example, Sentiment Look-Ahead [101] and RQL [100] condition responses to improve the user sentiment and adopt specific emotions and dialogue traits, respectively. Sentiment Look-Ahead adopts the MIXER sequence reinforcement strategy [138] with three possible sentiment reward functions. Meanwhile, RQL is a regularized Q-learning approach that selects from multiple candidate responses, based on the work by Zhou et al [21]. RQL proposes a custom-designed emotion aware reward function.

Another method for guiding text generation is to condition it on certain keywords. These are important to the perceived empathy of the text. For example, emotion cause words [56], [96], [106]. Emotion-cause keywords are used by Kim et al. [106] in a Bayesian framework to increase the probability of certain words being included in the generation. Similarly, Liu et al. [79] incorporate keywords as an input to a transformer decoder by using ConceptNet [84] and NRC_VAD [85]. Emp-RFT [96] obtains emotion and keyword pairs and uses them to recognize feature transitions. Subsequently, The next emotion and keywords are detected by Emp-RFT and an empathetic response is generated according to that information. A similar approach to Emp-RFT is ECTG by Qian et al. [31], which extracts emotion-cause keywords and uses them as vertices on a graph in which the edges are natural transitions between emotion

causes in the dialog. The ECTG graph is used to predict response concepts and fed to a Transformer decoder, which also uses a copy mechanism to ensure keywords are present in the response.

Conditioning on empathetic intent is a strategy taken by many articles. Xie and Pu [57], Chen et al. [16], and ConIntrEmp [103] use different methods to incorporate empathetic intent into the text generation pipeline with Transformers [66]. The most straightforward approach seems to predict an intent using a classifier and input the result to a Transformer decoder [57], [103]. However, EmpHi by Chen et al. models intent as a discrete latent variable given by an explainable CVAE (Conditional variational Autoencoder) [139]. EmpHi adds an emotional signal to the decoder and uses a copy mechanism that ensures intent keywords are present in the response. Intent conditioning and CVAEs are further used in hybrid models, as we will present in Subsection VII-C.

Explicit use of text responses for conditioning the model is seen in Learning to EMPathize using Exemplars (LEMPEx) by Majumder et al. [76], which append the fused representation of ''exemplar'' texts from the ED database. These texts are defined by being similar to the input context. Subsequently, this is fed to a Transformer decoder. A related approach by Fu et al. [74] uses an emotion classifier as a controlling signal: When the model is confident that an emotion is needed, a predetermined response from a database will be output. Pang et al. [73] present a rule-based approach: When the system detects that a user requires a validating response, two factors condition response generation (emotion and emotion cause). Then, predetermined formats are used to construct a response based on the confidence level of the emotion classifier. Finally, Sotolar et al. [88] propose using preference optimization and condition the responses by grounding them using completions with the same emotion and non-preferred completions with an opposite emotion.

Similarly, semantic relations can be used to explicitly control the generation of responses. We observe that in the ESCM [117] model, which builds semantic dependency trees that represent the correlation between emotions and semantics. These vectors are then applied to build a dynamic correlation graph convolutional network that guides the model.

Perhaps the most thorough method for variable conditioning is DiffusEmp [17]. To our knowledge, this is the only approach that uses Diffusion models for ETG. Diffusion models offer the possibility of explicit control of responses through signals. Bi et al. use a multi-grained control signal composed of intent [53], communication mechanism [32], and semantic frame [86] to have precise guidance over the response generation. A template response is obtained based on context and emotion similarity. Then, control signals are extracted from that template. Subsequently, a response is generated using the same parameters using the control signals.

A rarely explored variable conditioning approach is that of EmpSOI [99]. Wang et al. propose to obtain knowledge by building parallel knowledge graphs that permit the differentiation of both cognitive and affective cues from the user and the agent. Context relations and emotional knowledge are refined and fused using a gate control mechanism that adjusts the contribution of two perspectives to response generation.

Finally, we observe variable conditioning in LLM-based approaches. The PEGS LLM ETG system [18] is conditioned on Stickers present in the chat. It considers the input stickers, encodes them, and it generates responses with appropriate stickers in turn. Meanwhile, the Sibyl system [133] provides four commonsense knowledge categories as text: the underlying cause of the listener's next utterance, a possible subsequent event about the listener, the speaker's emotion state, and the intent of the speaker. The difference between these cues and those provided to the PECER [98] algorithm is that the prompt design of an LLM that uses the Sibyl system is instructed to explicitly incorporate those empathy cues into the response. As such, we consider it variable conditioning instead of context augmentation.

Conditioning on a given variable allows the models to obtain responses dependent on one or more empathy cues. As such, their performance goes beyond just acknowledgment and allows the systems to be more proactive. In addition, using variable conditioning seems to reduce the usage of generic responses by ETG systems [16], [31], [76]. As such, this is an effective method to enhance the perceived empathy of an ETG system, while ensuring that certain characteristics are present in the text.

Despite its effectiveness, we have identified some disadvantages with the variable conditioning strategy. For example, those that use text and keyword retrieval to condition a response present high computational complexity. Thus, higher latency. Additionally, variable conditioning approaches tend to focus on very few empathy cues. Therefore, they are focused on narrow aspects of empathy. For example, ETG algorithms that focus on intent and emotion can be limited by the number of labels available. Thus, negating other empathy cues from the cognitive dimension, like commonsense. There are very few variable conditioning approaches that use multiple control signals [17]. Empathy is a multi-dimensional concept. As such, considering multiple cues is important. To integrate diverse empathy cues, other approaches have attempted to combine variable conditioning and context augmentation.

C. HYBRID

The hybrid approach is the final strategy for integrating empathy cues seen in our survey. These are ETG systems that aim to combine the advantages of context augmentation and variable conditioning. To do this, most articles apply conditioning over a control signal and incorporate other empathy cues in their input.

One of the first hybrid approaches is the EmpDG proposed by Li et al. [78]. Coarse and fine-grained emotions are encoded using a transformer encoder. The resulting embeddings, alongside semantic representations, are used by the model to predict an emotion label that will be used to guide an empathetic response. Additionally, two discriminators (semantic and emotional) provide signals during training to ensure that the model has semantic and emotional guidance for the target response.

KEMP [80], CASE [90], and EMO_SA [83] are hybrid methods similar to EmpDG which use external databases to add commonsense and emotional knowledge into ETG. They construct context graphs from dialogue history. Subsequently, an encoder processes the graph vertices (words and concepts) and obtains emotion signals to guide response generation. In addition to the emotional context graph, EMO_SA incorporates the emotional tendency. The hybrid classification comes from integrating knowledge into the input (emotional context graph and emotion tendency) and using a guiding emotional signal for generating empathetic text.

Gao et al. [94] present a hybrid approach that includes emotion labels and emotion-cause. Emotion labels are integrated into the generation process through emotion embeddings. Furthermore, emotion-causes are used to guide the generation process. This is done by forcing the Transformer to pay attention to the emotion-cause words using soft and hard gates.

Other types of controlling signals besides emotions can be used by hybrid approaches. Similarly to variable condition, intent is a popular empathy cue for conditioning responses [52], [104], [112]. In addition, dialogue strategies [81] and the EPITOME communication mechanism [32] have been explored by ETG algorithms.

Liu and Kilicoglu [81] propose a hybrid modular ETG algorithm that uses plug-and-play components. They extract text prompts from the dialogue history using an external database and input them into a pre-trained BART text generation algorithm [129]. In addition, they obtain a controlling signal that represents a dialogue strategy [105]. In addition, a future discriminator enforces the strategy control. A remarkable attribute of the algorithm proposed by Liu and Kilicoglu [81] is the fact that it works with or without fine-tuning a pre-trained model, showing the feasibility of empathetic cues in ETG without having to retrain or fine-tune language models.

Gao et al. [52] propose Empathetic Dialogue Generation model including aspects of Cognition, Affection and Behavior (CAB). This model is one of the few designed to address the three dimensions of empathy. CAB incorporates knowledge through the cognitive and behavioral dimensions by enriching the context through ConceptNet [84] and an intent classifier [53]. The behavioral and cognitive variables are fed to the decoder. Regarding the Affective dimension, Gao et al. use two latent variables in a Transformer-based CVAE: Speaker emotion, and Listener emotion. Both variables are used to guide the model.

EPIMEED and EPIMEED+ [104] use intent and the EPITOME framework to guide the response generation process, while also addressing emotion. Meanwhile, EmpCI [112], a more recent approach proposed by Wang et al., uses emotion and intent as control signals in a two-step process, while also introducing commonsense knowledge to enrich the dialogue context.

Some ETG algorithms that use reinforcement learning perform hybrid strategies by conditioning responses on certain variables, while also performing context augmentation. RCLA [114] defines their actions as generating a response, this makes it so that the action space is effectively infinite. As such, they take an approach similar to previous methods, in which the model is first pre-trained to generate coherent responses [100], [101]. Then, they use a reward function that uses the EPITOME mechanisms and an anticipated emotion reward to ensure that the model displays higher emotional intelligence and empathy-rich responses. Context is augmented by extracting commonsense knowledge from the COMET model [116]. In contrast, EmpCRL [109] is an LLM-based model that uses emotion flow prediction and commonsense information as context augmentation. Additionally, a reinforcement learning algorithm is used to train an empathy driver that allows the model to control and match the emotions expected from the emotion flow.

The main advantage of using a hybrid strategy is that ETG algorithms are able to incorporate multiple empathy cues into their generation pipeline. Therefore, they represent more accurately the definition of empathy, as it is a multi-component and multi-dimensional concept [6], [11]. Furthermore, this strategy permits researchers to adopt previously explored cues while also contributing novel ones. We observe that there are less articles that propose a hybrid approach when compared to variable conditioning and context augmentation. Likewise, the cues explored are less diverse, with emotion and Commonsense knowledge being the most explored. Nevertheless, we propose that hybrid approaches can be considered the most promising methods for improving empathy performance using cues.

VIII. EVALUATING EMPATHETIC TEXT GENERATION SYSTEMS

Evaluating and comparing different ETG systems is challenging due to the subjective nature of empathy [12]. Therefore, identifying the best-performing ETG algorithms requires a novel evaluation methodology. In this section, we present the different methods used in the field to evaluate the performance of ETG systems. Likewise, we present a framework for selecting the SOTA models based on automatic evaluation.

For most approaches, the ED dataset [13] has been selected as a human baseline. However, we have identified that many of the entries display low empathy [61]. As such, this field requires a more scrutinized set of empathetic interactions to serve as a baseline. Nevertheless, ED is so far the most popular method. As such, it should

TABLE 5. Empathetic text generation models in our survey, grouped by integration strategy. M: Manual Annotation. C: Classification. R: Retrieval. G: Generative models.

Model	Year	Empathy Cues	Integration Strategy	M	C	R	G
XiaoIce (antecedent) [20]	2018	XiaoIce empathy vector	Context augmentation		✓		
EMC (antecedent) [21]	2018	Emotion	Context augmentation	✓			
EmoPrepend [13]	2019	Emotion	Context augmentation		✓		
TopicPrepend [13]	2019	Topic	Context augmentation		✓		
Multi-TRS [13]	2019	Emotion, Topic	Context augmentation		✓		
CAiRE [75]	2019	Emotion, Persona	Context augmentation	✓			
MoEL [15]	2019	Emotion	Context augmentation		✓		
EmpTransfo [30]	2020	Emotion	Context augmentation	✓			
EmoAwareTrs [93], [107]	2021	Emotion	Context augmentation		✓		
CEM [95]	2021	Commonsense, Emotion	Context augmentation		✓		✓
SEEK [24]	2022	Commonsense, Emotion, Intent	Context augmentation		✓		✓
EmpGPT-3 [51]	2022	Emotional situations	Context augmentation			✓	
EmpSOA [108]	2023	Commonsense, Emotion	Context augmentation				✓
EDG_MultiEmoInt [91]	2023	Emotion, Intent	Context augmentation			✓	✓
PEGS [18]	2024	Stickers	Context augmentation				✓
PECER [98]	2024	Commonsense, Emotion, Persona	Context augmentation		✓		✓
EmpDG [140]	2019	Emotion	Variable conditioning			✓	
MIME [14]	2020	Emotion, Sentiment	Variable conditioning		✓		
EC framework [106]	2021	Emotion-cause	Variable conditioning				✓
EMMA [56]	2021	Emotion	Variable conditioning	✓			
RGPT2 + CKECE [79]	2021	Emotion	Variable conditioning			✓	
Sentiment Look-Ahead [101]	2021	Sentiment	Variable conditioning		✓		
MEED [57]	2021	Emotion, Intent	Variable conditioning		✓		
Gao et al. [94]	2021	Emotion-cause	Variable conditioning		✓		
EmpHI [16]	2022	Emotion, Intent	Variable conditioning		✓		
Emp-RFT [96]	2022	EPITOME, Emotion, Emotion-cause, Keywords	Variable conditioning		✓		✓
LEMPEx [76]	2022	EPITOME, Emotion, Exemplar, Sentiment	Variable conditioning		✓	✓	
DiffusEmp [17]	2023	EPITOME, Emotion, Intent, Semantic, Candidates	Variable conditioning		✓	✓	
Fu et al. [74]	2023	Emotion, Exemplars	Variable conditioning	✓	✓	✓	
ECTG [31]	2023	Emotion-cause	Variable conditioning	✓			✓
ConIntrEmp [103]	2023	Emotion, Intent	Variable conditioning		✓		
ESCM [117]	2023	Emotion-Semantic	Variable conditioning				✓
Validating responses [73]	2024	Emotion, Validation	Variable conditioning	✓	✓		
InfRa [118]	2024	SDRT	Variable conditioning				✓
RQL [100]	2024	Dialogue Traits, Emotion, Sentiment	Variable conditioning		✓		
EmPO [88]	2024	Preferred and non-preferred completions	Variable conditioning			✓	
EmpSOI [99]	2024	Commonsense, Emotion, Perspective	Variable conditioning		✓		✓
+ Sibyl [133]	2025	Cause, Subsequent event, Emotion state, Intent	Variable conditioning				✓
KEMP [80]	2022	Emotion	Hybrid			✓	
EMO_SA [83]	2023	Emotion	Hybrid			✓	
CAB [52]	2023	Commonsense, Emotion, Intent	Hybrid		✓	✓	
CKEDG [81]	2023	Prompts, dialogue strategy	Hybrid			✓	✓
EPIMEED, EPIMEED+ [104]	2023	EPITOME, Emotion, Intent	Hybrid		✓		
CASE [90]	2023	Commonsense, Emotion, Keywords	Hybrid		✓	✓	✓
RLCA [114]	2024	Commonsense, EPITOME, Emotion	Hybrid		✓		✓
EmpCRL [109]	2024	Commonsense, Emotion	Hybrid		✓		✓
EmpCI [112]	2024	Commonsense, Emotion, Intent	Hybrid		✓		✓

be considered in experimentation. Otherwise, alternative similar datasets could used, like EDOS [55] or JPED [72]. Additionally, we suggest that future research utilize datasets containing verified empathetic interactions. For example, EmpatheticExchanges [11] provides human annotated labels for perceived empathy for a subset of ED. Interactions that display empathy according to manually annotated empathy labels could serve as a reliable test set for ETG systems.

The automatic evaluation metrics observed in our survey show that there is no consensus on how to accurately evaluate empathy [13]. The most popular metrics have been Perplexity (PPL), Distinct-n (DS-n), BERTscore, and BLEU [141]. PPL measures how well a probabilistic model predicts a given sample [141]. However, in ETG, PPL has served as metric to measure the fluency and quality of a generated response [51], [98]. Meanwhile, relevance of generated text has been obtained using the BERTscore, which measures the semantic similarity between generated responses and golden references [142]. Furthermore, Distinct-n, a metric for the distinct n-grams in a generated response [143], has been used to measure the diversity and the informativeness of responses [17]. The final metric, BLEU, has been proposed for measuring the informativeness of the generated responses. However, this metric and ROUGE [144] should not be considered as appropriate metrics, as they do not correlate to human judgment [145]. PPL, DS-n, and the BERTscore can serve as starting points towards automatic evaluation of ETG

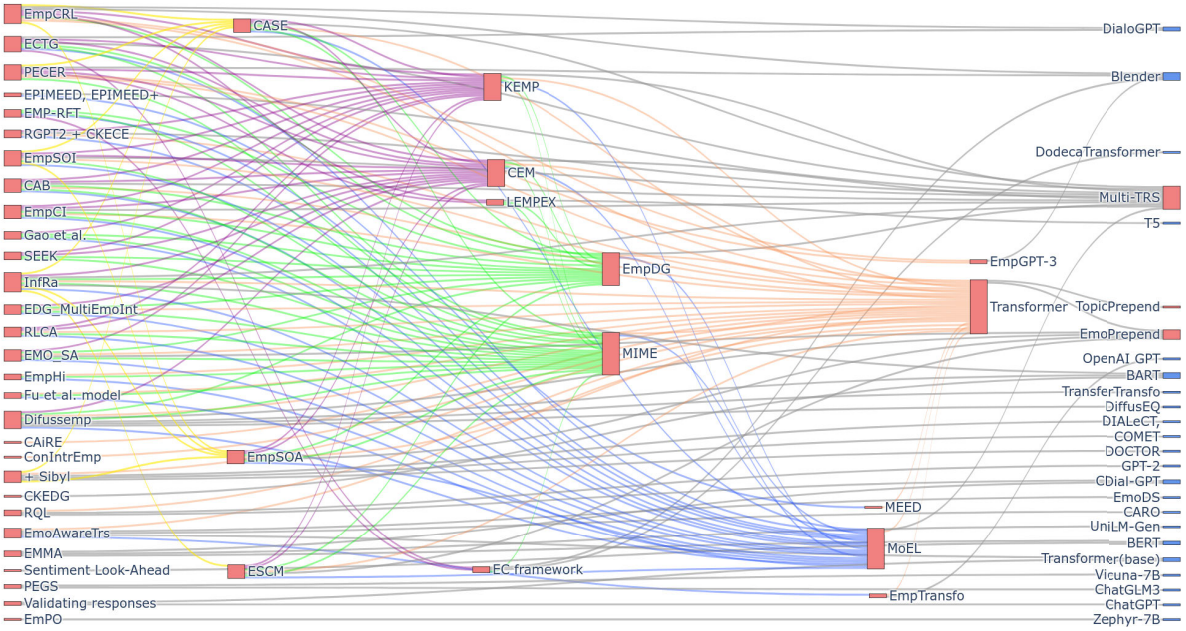


FIGURE 3. Hierarchical comparison between models surveyed. Red nodes represent models designed for ETG, whereas blue nodes are other text generation models. Each link represents the model on the left outperforming the model on the right using the same dataset and metrics. Gray links: Model outperforms a baseline model, which has not outperformed any other. Orange links: Model outperforms another model that has previously outperformed a baseline, the model on the left is 1 comparison level above baseline models. Blue links: Victory over a model that has outperformed models shown to be superior to baselines, the model on the left 2 depth levels above the baseline. Green links: The model on the left is 3 comparison levels above baseline models. Purple links: Model is 4 hierarchical comparison levels above baseline models. Yellow links: Model outperforms another model at 4 comparison levels, the model on the left is at the top hierarchical comparison between models.

systems. However, empathetic performance must be obtained using different metrics.

Reliable automatic metrics for empathy have yet to be developed. Nevertheless, we observe some approaches based on the EPITOME framework [32]. This framework measures the intensity of specific communication mechanisms in dialogue: Emotional Reaction (EM), Interpretation (IP), and Exploration (EX). These were proposed by Sharma et al. to be characteristics that describe empathetic communication. Therefore, they work as empathy metrics. Lee et al. [51] further develops the EPITOME framework as an empathy metric by proposing Diff-EPITOME a multi-dimensional metric that measures the difference scores of IP, EX, and ER between the human golden response and a predicted response. Further work should be carried out in this area. However, many proposals have been made towards automatic detection of empathy in text [11], [146]. Adopting another metric, such as labels from empathy classifiers, could be an alternative method to obtain an automatic empathy metric for ETG.

Human Evaluation has also been observed in our survey. Mostly, studies follow the same metrics as Rashkin et al. [13]: Fluency, Empathy, and Relevance in a five-point scale. However, additional ones like Coherence, Informativeness, and Specificity have also been proposed [31]. Human evaluation can also be performed by exploring the preference of human annotators in regards to the metrics previously stated [98], rather than evaluating them using a scale. However, we observe this method to be less common. Empathy is difficult to automatically classify and measure in text [12], [146]. Therefore, human evaluation continues to be necessary for any ETG system. Nevertheless, we develop our framework for ranking ETG models based on automatic evaluation, as human evaluation lacks a consistent methodology.

Selection of benchmarks should be carried out while considering the SOTA. We provide an overview of the models surveyed according to their performance in Figure 3. For this overview, we provide a framework in which we compare model performance based on automatic evaluation metrics using the same datasets. We consider the individual research articles of each model and obtain the models used as benchmarks. Then, we graph all comparative victories of each model against others. Each comparative victory is obtained in experiments with the same dataset and metrics. However, they do not share collectively the same parameters. We consider that a model outperforms another when it obtains better results than its adversary in the majority of metrics.

Thanks to this framework, we can obtain a hierarchical view of performance across the field. Each comparison reflects the model outperforming another using the same samples. We include models not designed specifically for ETG in our analysis, as they have been compared to ETG systems in various articles. Additionally, we provide a full list of comparisons in Table 6.

TABLE 6. Comparisons between models and the adversaries they outperform. Outperforming the models refers to them obtaining better results in the majority of their metrics with the same dataset.

Model	Models outperformed
Transformer [13], [66]	EmoPrepend [13] Topic [13]
CAiRE [75]	Transformer [13], [66]
MoEL [15]	Transformer [13], [66], Multi-TRS [13],
EmpDG [89]	Transformer [13], [66], Multi-TRS [13], MoEL [15]
MIME [14]	Transformer [13], [66], MoEL [15]
EmpTransfo [30]	Transformer [13], [66], OpenAI GPT [147]
EmoAwareTrs [93], [107]	Transformer [13], [66], CARO [148], Transformer [66], EmoDS [149], EmpTransfo [30]
Gao et al. [94]	EmpDG [89],MoEL [15], MIME [14], KEMP [80]
EC framework [106]	MIME [14], DodecaTransformer [130], Blender [150]
EMMA [56]	BERT [121], UniLM-Gen [151], CDial-GPT [152]
RGPT2 + CKECE [79]	Transformer [13], [66], Multi-TRS [13], MoEL [15], KEMP [80]
Sentiment Look-Ahead [101]	BERT [121]
MEED [57]	Transformer [13]
CEM [95]	Transfomer [13], [66], Multi-TRS [13], MoEL [15], MIME [14], EmpDG [89]
KEMP [80]	MoEL [15], Transfomer [13], [66], Multi-TRS [13], MIME [14], EmpDG [89]
EmpHi [16]	Transfomer [13], [66], MoEL [15], MIME [14]
EMP-RFT [96]	MoEL [15], MIME [14], EmpDG [89], EC framework [106]
LEMPEX [76]	MIME [14], EmpDG [89], T5 [134]
SEEK [24]	MIME [14], EmpDG [89], KEMP [80], CEM [95]
Difussemp [52]	Transformer [13], [66], EmoPrepend [13], MoEL [15], MIME [14], EmpDG [89], CEM [95], BART [129], TransferTransfo [153], DiffusEQ [154]
EMO_SA [83]	KEMP [80], MoEL [15], Transfomer [13], [66], EmoPrepend [13], MIME [14], EmpDG [89]
Fu et al. model [74]	Transformer [13], [66], MoEL [15], MIME [14]
CAB [52]	Transformer [13], [66], Multi-TRS [13], MoEL [15], MIME [14], KEMP [80], EmpDG [89], CEM [95]
CKEDG [81]	BART [129]
ECTG [31]	Multi-TRS [13], MoEL [15], MIME [14], EmpDG [89], EC framework [106], KEMP [80], CEM [95], DialoGPT [155]
ConIntrEmp [103]	Transformer [13], [66]
EmpSOA [108]	Transformer [13], [66], EmoPrepend [13], MoEL [15], [14], EmpDG [89], KEMP [80], CEM [95]
EPIMEED, EPIMEED+ [104]	Multi-TRS [13], MEED [57]
PEGS [18]	Vicuna-7B [123], ChatGLM3 [156]
Validating responses [73]	Transformer [66], ChatGPT
EDG_MultiEmoInt [91]	Transformer [13], [66], MoEL [15], MIME [14], EmpDG [89], CEM [95]
ESCM [117]	Transformer [13], [66], EmoPrepend [13], MoEL [15], MIME [14], EmpDG [89], KEMP [80], CEM [95]
RQL [100]	Transformer [66], GPT-2 [152] , CDial-GPT [152]
InfRa [118]	Transformer [66], Multi-TRS [13], MoEL [15], MIME [14], EmpDG [89], CEM [95], CASE [90], ESCM [117], EmpSOA [108], BART [129]
EmpGPT-3 [51]	Blender [150]
EmpCRL [100]	Transformer [13], [66], EmpDG [89] , KEMP [80], CEM [95], CASE [90], EmpSOA [108], Blender [150], DialoGPT [155], LEMPEx [76], EmpGPT-3 [51]
CASE [90]	Transfomer [13], [66], Multi-TRS [13], MoEL [15], MIME [14], EmpDG [89], KEMP [80], CEM [95]
EmpCI [112]	Transfomer [13], [66], Multi-TRS [13], MoEL [15], MIME [14], EmpDG [89], CEM [95]
PECER [98]	Transfomer [13], [66], MIME [14], KEMP [80], CEM [95], CASE [90], LEMPEX [76], Blender [150], EmpGPT-3 [51]
RLCA [114]	Transfomer [13], [66], MoEL [15], MIME [14], KEMP [80], CEM [95]
EmPO [88]	Zephyr-7B
EmpSOI [99]	Transfomer [13], [66], Multi-TRS [13], MoEL [15], MIME [14], KEMP [80], CEM [95], CASE [90], EmpSOA [108]
+ Sibyl [133]	Transfomer [13], [66], CASE [90], +COMET [133], +DOCTOR [133], +DIALeCT [133], EmpSOA [108]

We consider baseline models those that have not being able to outperform any other in the ETG task. Most of them were designed for different purposes. However, other models proposed by Rashkin et al. [13] (Multi-TRS, TopicPrepend, and EmoPrepend) are in the baseline category. A level above this category, we find the Transformer [13], [66]

and EmpGPT-3 [51] models, which have managed to outperform baselines. The Transformer model trained on ED has been extensively used as a point of comparison in ETG. As such, it is an adequate basic benchmark for ETG system. Nevertheless, other popular models like MoEL [15], EmpTransfo [30] and MEED [57] have been used as benchmarks in various articles, while having outperformed the Transformer model alongside other baselines. Subsequently, we find EmpDG [89] and MIME [14] a level above those previously discussed, with them obtaining victory over MoEL specifically. Then, we observe KEMP [80], CEM [95], LEMPEX [76], and the EC framework [106], a hierarchical level above EmpDG and MIME. CASE [90], EmpSOA [108], and ESCM [117] are one of the highest ranked ETG algorithms, managing to outperform those on the previous level, but still being surpassed by models like EmpCRL [109], InfRa [118], + Sibyl [133], PECER [98], and EmpSOI [99]. The latter group are what we consider the most appropriate benchmarks up to now to compare a novel algorithm. However, we believe other algorithms on an immediate lower level can also serve as supplementary benchmarks. Finally, it is important to note that the most popular benchmark models are KEMP, CEM, EmpDG, MIME, MoEL, and the Transformer. Thus, we recommend using only one or two of these models during testing, on account of their low performance compared to other systems.

We observe that newer models that use the variable conditioning strategy, either alone or as a hybrid strategy, such as InfRa [118], + Sibyl [133], EmpSOI [99], and EmpCRL [109], perform better in the ETG task than those approaches that use context augmentation, with the only exception being PECER [98], an LLM-based system. Likewise, thank to our analysis we can state that the Generative Models sourcing method is the most common in better-performing. Therefore, the type of cues extracted through this method should be considered for future appproaches.

IX. DISCUSSION

In this survey, we presented multiple empathetic text generation (ETG) systems and analyze how they use variables to improve the perceived empathy in text. We defined these variables as empathy cues, as they can be used to align a text towards the affective (emotion, sentiment, emotion reaction, emotion cause), cognitive (feature transition, commonsense, interpretation, exploration), or behavioral (intent, keywords, stickers, validation) dimensions of empathy.

The trend observed in datasets is that there was a surge of development after EMPATHETICDIALOGUES (ED) [13] became public in 2019. ED is still one of the datasets most used for this purpose, alongside its variations [53], [72]. However, to use empathy cues in this dataset beyond the manually annotated emotions, it is necessary to expand and augment the data using specific methodologies, such as manual annotation followed by classification [53], [73]. The emotions in the database are one of the empathy cues most used by the algorithms explored [16], [73], [74], [96]. However, concerns have been raised on whether their labels are accurate [60]. Further work must be carried out to obtain databases that are as flexible as ED but address its shortcomings.

The methods used to extract and integrate empathy cues in ETG have evolved rapidly since ED was released. Likewise, diverse empathy cues beyond emotion labels have been proposed since. Consequently, many sources and algorithms have been leveraged to use them in ETG. We identified the methods for sourcing these cues from a dialogue history: When a cue is first proposed, researchers tend to define and manually annotate empathy components. This is the only strategy that requires human labor and it is the basis for developing empathy cue datasets. Subsequently, three automatic labeling methods can be used to include them in the text generation pipeline: classification, retrieval from an external database, and through generative models. Sourcing methods translate empathy components into useful numerical, categorical, and textual values. Separating these methods into categories allowed us to differentiate cue sources across the field of ETG. We presented the different cues that can be sourced through each of the categories and described their advantages and possible disadvantages.

We observed that once cues are obtained, they can be leveraged by ETG systems to increase the perceived empathy of a response. It is important to note that the nature of the cues ties it heavily to the sourcing method. For example, labels like emotion and sentiment tend to be extracted through classification algorithms [13], [16], [74], whereas sequential data uses generative models [31], [94], [96], [106]. This means that models with more than one empathy cue usually require multiple extraction methods.

We presented strategies for implementing empathy cues into the text generation pipeline. We proposed three categories: context augmentation, variable conditioning, and hybrid strategies. Each of these methods have grown in complexity over time. The first ETG approaches performed context augmentation through empathy cues without without any changes to internal mechanisms [13], [30]. Subsequently, the use of control variables has become much more popular. This is particularly true for empathy cues related to behavioral empathy [16], [31], [57], [96], [103], [106]. Finally, we saw the most advanced methods in hybrid approaches. They aim to address multiple empathy cues in their context and use control signals to leverage the advantages of both previous approaches.

In our survey we observed that models adopt different integration strategies depending on the method used to source empathy cues.

Manually annotated empathy cues, mainly emotion and emotion-cause, were initially found to be integrated through context augmentation. Then, cues were integrated using variable conditioning, such as the case with exemplars and validation timing. These cues are entirely missing from hybrid methods explored in this survey. Therefore, novel cues

that require annotating could be integrated through a hybrid strategy in the future.

Cues obtained through retrieval are primarily integrated using variable conditioning and hybrid strategies. For example, lexicons and semantic knowledge can be used to construct emotional relations to enrich the context of an input. Meanwhile, exemplars and similar situations from a database can be integrated through variable conditioning and hybrid strategies to encourage human-like responses.

Most ETG algorithms, regardless will adopt cues sourced from classification algorithms. Therefore, these cues present the most diversity in integration strategies. Emotion labels can be used for context augmentation, this is observed with those proposed by Rashkin et al. [13]. Meanwhile, empathetic intent, the second most popular cue obtained through classification, is heavily used as a control signal in variable conditioning and hybrid strategies.

Complex cues like those obtained from generative models are likewise compatible with all integration strategies. However, we see them being adopted later in the timeline compared to those obtained through classification. Language models are popularly used to extract commonsense knowledge, integrated largely through the context augmentation strategy. In contrast, cues integrated through the variable control strategy are in the form of keywords and dialogue characteristics. These cues control the nature of a response based on the output of a supplementary model.

The field of empathetic computing is developing rapidly, with many articles published constantly. Therefore, acknowledging the trends in which studies are heading is important. Currently, there is a very scattered approach that leverages different algorithms. Nevertheless, we see connections between articles that could help solidify the direction to take this research. Our analysis shows that hybrid approaches and combining different empathy cues present potential for improving ETG performance.

A. AREAS OF OPPORTUNITY

In this subsection, we present possible areas of opportunity that future researchers can explore. There is potential in this field to improve ETG and affective computing in general through exploration of empathy cues and their connection through psychological concepts. Likewise, using different cues combined with previously proposed architectures can lead to a more cohesive development of this field.

Regarding databases, it is still necessary to carry out studies that verify the reliability of the most popular databases. For example, ED is still used heavily in ETG despite containing questionable variables. Research to verify other databases related to empathetic dialogues is necessary. Likewise, testing existing ETG systems in other datasets has the potential of yielding insight into the application of these proposals.

We propose that an effort is made to augment or generate new databases with reliable empathy, and empathy cue annotations. This approach has the potential to produce a new baseline dataset with verified empathy responses that also standardizes empathy cues. For this, the most popular empathy cues could be manually annotated. For instance, intent, emotion, the EPITOME framework, commonsense knowledge and emotion-cause.

Another avenue of opportunity lies in using LLMs to obtain and augment databases. This is a viable strategy in previous research [18]. Therefore one research gap to be explored is to compared the performance of LLMs to manual annotation and determine if using an automatic system could yield reliable datasets for ETG.

Regarding manually annotated empathy cues, we recommend to focus on taking advantage of the reliability of human evaluation and develop novel empathy cues. In particular, focus on obtaining taxonomies and labels that could later be predicted automatically. Integrating them with variable conditioning can enhance empathetic performance without the uncertainty of automatic sources like classification. Likewise, they can be integrated into LLMs and traditional text generation pipelines by using context augmentation to gauge their influence without algorithm modification. Finally, testing manually annotated cues with hybrid integration strategies is still an open avenue to explore.

Regarding empathy cues obtained from retrieval, we observed that there is potential to compare them to their equivalents from other sourcing methods. For example, lexicons allow for extracting emotion through words and it has been shown to be more reliable in empathy detection than using classifiers [11]. Exploring the impact of retrieval-sourced emotion compared to emotion labels in ETG could be an approach to gauge whether a cue obtained from retrieval is a more reliable method than machine learning approaches for increasing empathetic performance.

For classification cues, there are very few approaches that use a different classification taxonomy for emotion and intent. Therefore, future research could focus on whether the number of emotion labels is optimal. The 32 emotion labels have been extensively applied in ETG. However, a reduced number of labels might perform better. Likewise, studies could be carried out to explore whether empathetic intent could be replaced by similar concepts as a control signal. For example, the dialogue strategy or the EPITOME framework.

Generative models can obtain complex empathy cues. However, we see that commonsense knowledge through the COMET model is overwhelmingly used. Alternatives like LLMs can be used to extract empathy cues from psychological research, such a specific strategies and characteristics of empathetic communication [6]. Likewise, other types of commonsense knowledge beyond the COMET model could be tested using the same architecture to gauge the differences between those cues.

It would be beneficial if future articles separate empathy cues by their empathy dimension: Whether the cue can be classified as part of affective, cognitive, or behavioral dimension. In this manner, it would be easier to identify which cues and sourcing methodologies align more with

the dimensions in psychological research. Furthermore, more ETG systems must explicitly develop top-down approaches based on cues from psychological research. Particularly, we believe using behavioral empathy components as control signals could improve perceived empathy.

Regarding empathy cue integration, there is a trend towards hybrid approaches. We believe that this is reasonable since empathy is a multi-dimensional concept. Nevertheless, work must be done to compare the benefits of using multiple empathy components in hybrid approaches. For example, comparing intent and keyword conditioning could yield results that tell researchers which is more effective. This is to discourage adding empathy cues without a strong theoretical basis. Furthermore, we have not observed papers discussing the cost-benefit analysis for complexity in different hybrid models.

We see ETG systems implementing multiple empathy cues (e.g. emotion classification, intent classification, and emotion cause). However, we do not observe that the architectures explored in this survey built upon each other. Most ETG system propose novel approaches for text generation with unique cues. As such, there are research gaps regarding whether combining the different approaches in a single architecture can improve ETG. Therefore, we opine that pursuing multiple already explored empathy cues into a single ETG system could offer benefits. A single system could be developed using LLMs to easily integrate many cues simultaneously, as these models have proven to be very versatile. Nevertheless, this also requires refinement of empathy sourcing methods, to make those algorithms used more accurate and efficient. Likewise, researchers should ensure that cues are not added unnecessarily.

Further work has to be carried out to standardize evaluation in this field. We have found a diverse group of automatic and human metrics that are completely incompatible. As such, it has been challenging to ascertain the quality of different algorithms. We propose that future research uses metrics based on the perceived empathy of humans. This can be done through metrics like the EPITOME framework [32], [76], [106]. Alternatively, a scale for perceived empathy [61] could be used. Perceived empathy scales were used for human evaluation in many articles [13], [17], [76], but they also have been explored in classification algorithms [61]. Another option is to develop a new metric, which could be done based on the work by Brannstromm et al. [9]. Likewise, selecting models as standard benchmarks could allow ETG systems to build upon each other and create a more organized field. We provide our approach for selecting benchmarks and expect that future research builds upon it.

Some Bayesian algorithms have been used in ETG to control text generation using empathy cues. We propose that further work continues to explore these approaches. Empathy is a complex concept. Therefore, explainability must be considered when developing ETG systems to provide insight into this concept in communication. Using latent variables in Bayesian algorithms allows researchers to interpret the importance of variables in ETG. As such, we believe this is an important area of opportunity. Likewise, ETG systems that use LLMs could provide their reasoning for their responses, which could provide explainability.

Finally, the work by Gao et al. [52] has shown that interpreting empathy cues through the dimensions of empathy is a valuable method to improve ETG. We propose that future approaches follow this method, proposing hybrid approaches that use context augmentation on cognitive and affective components while using behavioral components as a conditioning variable.

X. CONCLUSION

We carried out a survey to identify the methods used to create empathetic text responses in one-on-one open-domain conversations. Our survey focused on the variables related to empathy that we denominate *empathy cues* and how they can be leveraged for increased perceived empathy in text. The methods for extracting empathy cues and integrating them into text generation were represented in a novel framework. Four methods were observed for obtaining empathy cues. Meanwhile, 3 were discerned for implementing them into the text generation pipeline. We identified the most popular evaluation metrics an obtained a hierarchical overview of performance across the field. Likewise, we presented a trend towards more complex hybrid systems overtime since the start of this field. Finally, we identified areas of opportunity that can be useful for future research.

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